

Numerical Exercise for Schedule and Cost Metrics

1. A project has a budget of \$200,000 and is scheduled to finish in 20 weeks. By the end of week 10:

- Planned progress = 60% of total work
- Actual progress = 50% of total work

Calculate PV, EV, SV, and SPI.

2. A project has a budget of \$150,000 and is scheduled to finish in 15 months. By the end of month 9:

- Planned progress = 70% of total work
- Actual progress = 75% of total work

Calculate PV, EV, SV, and SPI.

3. A software development project has a total budget of \$200,000 and is scheduled to finish in 20 weeks. The project manager wants to evaluate schedule performance at different checkpoints using Planned Value (PV), Earned Value (EV), Schedule Variance (SV), and Schedule Performance Index (SPI).

By monitoring progress at three checkpoints, the following data was collected:

- Checkpoint 1 (Week 6): Planned progress = 40% Actual progress = 35%
- Checkpoint 2 (Week 10): Planned progress = 60% Actual progress = 50%
- Checkpoint 3 (Week 15): Planned progress = 70% Actual progress = 75%

For each checkpoint, calculate:

1. Planned Value (PV)
2. Earned Value (EV)
3. Schedule Variance ($SV = EV - PV$)
4. Schedule Performance Index ($SPI = EV \div PV$)

4. A construction project planned 20 milestones for Phase 1. By the review date, 16 milestones were completed on time. The critical path was scheduled for 45 days, but actual completion took 50 days. Three tasks had cycle times: Task A = 6 days, Task B = 8 days, Task C = 7 days. A client request was submitted on April 1 and delivered on April 18. Task delays: Task 1 (planned 10 days, actual 12), Task 2 (planned 8 days, actual 9), Task 3 (planned 6 days, actual 8).

Calculate:

- On-Time Completion Rate
- Critical Path Variance

- Average Cycle Time
- Lead Time
- Average Task Delay

5. A sprint planned 15 milestones, and 12 were achieved on time.

- The critical path was scheduled for 30 days, but actual completion took 28 days.
- Cycle times for features: Feature X = 5 days, Feature Y = 7 days, Feature Z = 6 days, Feature W = 4 days.
- A bug fix request was submitted on June 10 and delivered on June 22.
- Task delays: Task A (planned 7 days, actual 9), Task B (planned 5 days, actual 6), Task C (planned 9 days, actual 11), Task D (planned 4 days, actual 5).

Calculate:

- On-Time Completion Rate
- Critical Path Variance
- Average Cycle Time
- Lead Time
- Average Task Delay

6. A campaign planned 25 milestones, and 20 were completed on time.

- The critical path was scheduled for 60 days, but actual completion took 65 days.
- Cycle times for deliverables: Deliverable A = 10 days, Deliverable B = 12 days, Deliverable C = 8 days.
- A client request was submitted on September 5 and delivered on September 25.
- Task delays: Task 1 (planned 12 days, actual 15), Task 2 (planned 9 days, actual 11), Task 3 (planned 6 days, actual 7), Task 4 (planned 10 days, actual 13).

Calculate:

- On-Time Completion Rate
- Critical Path Variance
- Average Cycle Time
- Lead Time
- Average Task Delay

7.

A software module is planned to cost ₹500,000 and take 10 weeks. After 5 weeks, the following data is collected:

- Planned Value (PV): 60% of work should have been completed.
 - Earned Value (EV): Only 50% of work is actually completed.
 - Actual Cost (AC): ₹280,000 has been spent so far.
- a. Compute PV in rupees.
 - b. Compute EV in rupees.
 - c. Determine the Cost Variance ($CV = EV - AC$).
 - d. Calculate the Cost Performance Index ($CPI = EV \div AC$).
 - e. Interpret whether the project is under or over budget.

8.

A foundation-laying activity is budgeted at ₹1,200,000 for the first phase. After 30 days, the progress shows:

- PV: ₹900,000
 - EV: ₹750,000
 - AC: ₹820,000
- a. Verify whether the project is ahead or behind schedule using PV vs EV.
 - b. Compute Cost Variance ($CV = EV - AC$).
 - c. Compute CPI ($EV \div AC$).
 - d. State whether the project is spending efficiently.
 - e. Briefly state what corrective action may be needed based on CPI.

9.

A company is upgrading its assembly line with a total budget of ₹2,500,000. At the mid-point review:

- Planned progress: 50%
 - Actual completed work: 55%
 - Actual Cost spent (AC): ₹1,600,000
- a. Calculate Planned Value (PV).
 - b. Calculate Earned Value (EV).
 - c. Compute $CV = EV - AC$.
 - d. Compute $CPI = EV \div AC$.
 - e. Interpret:
 - Is the project ahead or behind schedule?
 - Is it under or over budget?

10. A mobile-app development project has a Budget at Completion (BAC) of ₹2,000,000. So far:

- Actual Cost (AC) incurred = ₹1,200,000
- Earned Value (EV) = ₹1,000,000
- Cost Performance Index (CPI) = $EV \div AC$
- Cost of Quality (CoQ) spent so far = ₹150,000
- The project generated projected business benefits worth ₹3,500,000
 - a. Compute CPI.
 - b. Calculate the Estimate at Completion ($EAC = BAC \div CPI$).
 - c. Compute ETC = $EAC - AC$.
 - d. Compute Variance at Completion ($VAC = BAC - EAC$).
 - e. Compute ROI = $(Benefits - EAC) \div EAC$.

CPI

$$CPI = \frac{EV}{AC} = \frac{1,000,000}{1,200,000} = 0.83$$

EAC

$$EAC = \frac{BAC}{CPI} = \frac{2,000,000}{0.83} \approx ₹2,409,639$$

ETC

$$ETC = EAC - AC = 2,409,639 - 1,200,000 = ₹1,209,639$$

VAC

$$VAC = BAC - EAC = 2,000,000 - 2,409,639 = -₹409,639$$

→ Project is forecasted over budget by ₹409,639.

ROI

$$ROI = \frac{Benefits - EAC}{EAC} = \frac{3,500,000 - 2,409,639}{2,409,639}$$

$$ROI \approx 0.45 = 45\%$$

11. A housing foundation project has:

1. BAC = ₹8,000,000
2. Actual cost to date (AC) = ₹5,000,000
3. Earned Value (EV) = ₹4,400,000
4. Benefits expected from property sale = ₹11,000,000

5. Cost of Quality (CoQ) including rework = ₹360,000
6. Resource cost rate = ₹2,000 per labour hour
7. Labour used so far = 2,400 hours
 - a. Compute CPI.
 - b. Compute EAC (BAC ÷ CPI).
 - c. Compute ETC = EAC – AC.
 - d. Compute VAC = BAC – EAC.
 - e. Compute Labour Cost = Resource Cost Rate × Hours Worked.
 - f. Compute ROI = (Benefits – EAC) ÷ EAC.

CPI

$$CPI = \frac{EV}{AC} = \frac{4,400,000}{5,000,000} = 0.88$$

EAC

$$EAC = \frac{BAC}{CPI} = \frac{8,000,000}{0.88} = ₹9,090,909$$

ETC

$$ETC = 9,090,909 - 5,000,000 = ₹4,090,909$$

VAC

$$VAC = 8,000,000 - 9,090,909 = -₹1,090,909$$

→ Forecasted budget overrun: ₹1.09 million

Labour Cost

$$\text{Labour Cost} = 2,000 \times 2,400 = ₹4,800,000$$

ROI

$$ROI = \frac{11,000,000 - 9,090,909}{9,090,909}$$

$$ROI \approx 0.21 = 21\%$$

12. A company is upgrading its data-center servers.

- BAC = ₹5,500,000
- AC = ₹3,000,000
- EV = ₹3,300,000
- Projected business benefits = ₹8,000,000
- CoQ (testing + preventive checks) = ₹420,000
- Resource cost rate = ₹3,500 per engineer hour

- Hours consumed = 700 hours
 - a. Compute CPI.
 - b. Compute $EAC = BAC \div CPI$.
 - c. Compute $ETC = EAC - AC$.
 - d. Compute $VAC = BAC - EAC$.
 - e. Compute Resource Cost (Rate $\times$ Hours).
 - f. Compute ROI.

CPI

$$CPI = \frac{EV}{AC} = \frac{3,300,000}{3,000,000} = 1.10$$

→ A CPI greater than 1 indicates cost efficiency.

EAC

$$EAC = \frac{5,500,000}{1.10} = ₹5,000,000$$

ETC

$$ETC = 5,000,000 - 3,000,000 = ₹2,000,000$$

VAC

$$VAC = 5,500,000 - 5,000,000 = ₹500,000$$

→ Project is forecasted under budget by ₹500,000.

Resource Cost

$$\text{Resource Cost} = 3,500 \times 700 = ₹2,450,000$$

ROI

$$ROI = \frac{8,000,000 - 5,000,000}{5,000,000} = 0.60 = 60\%$$

Numerical Exercise for COCOMO & FP

1. A software project is estimated to be 32 KLOC and classified as an Organic type project.

Organic Mode Constants

- $a = 2.4$
- $b = 1.05$

- $c = 2.5$
- $d = 0.38$
 - Compute the Effort (E) in person-months.
 - Compute the Development Time (D) in months.
 - Compute the Average Staffing (P).

2.

A project is estimated at 50 KLOC and classified as a Semidetached project.

Semidetached Mode Constants

- $a = 3.0$
- $b = 1.12$
- $c = 2.5$
- $d = 0.35$

Effort Adjustment Factor (EAF)

Cost Drivers:

Cost Driver	Rating	Multiplier
RELY	High	1.15
DATA	Nominal	1.00
CPLX	High	1.15
TIME	Nominal	1.00
ACAP	Nominal	1.00
PCAP	High	0.88

- Compute EAF.
- Compute the Effort (E).
- Compute the Development Time (D).
- Compute Average Staffing (P).

3. A project is 20 KLOC and categorized as Semidetached.

Semidetached Constants

- $a = 3.0$

- $b = 1.12$
- $c = 2.5$
- $d = 0.35$
 - Compute Effort (E)
 - Compute Development Time (D)
 - Compute Average Staff (P)

4. A system software project (Embedded mode) is 60 KLOC.

Embedded Constants

- $a = 3.6$
- $b = 1.20$
- $c = 2.5$
- $d = 0.32$
 - Compute Effort (E)
 - Compute Development Time (D)
 - Compute Staffing (P)

5.

A 48 KLOC Semidetached software project has the following cost drivers:

Cost Driver	Rating	Multiplier
RELY	Nominal	1.00
DATA	High	1.14
CPLX	Nominal	1.00
ACAP	High	0.86
PCAP	Nominal	1.00
TIME	Low	1.00

Constants for Semidetached

- $a = 3.0$
- $b = 1.12$
- $c = 2.5$
- $d = 0.35$

- Compute EAF
- Compute Effort (E)
- Compute Development Time (D)
- Compute Staffing (P)

6.

A 75 KLOC Embedded project has these cost drivers:

Cost Driver	Rating	Multiplier
RELY	High	1.15
CPLX	Very High	1.30
ACAP	Nominal	1.00
PCAP	High	0.88
TIME	High	1.10
STOR	Nominal	1.00

Embedded Constants

- $a = 3.6$
- $b = 1.20$
- $c = 2.5$
- $d = 0.32$
- Compute EAF
- Compute Effort (E)
- Compute Development Time (D)
- Compute Staffing (P)

7. A small retail shop needs a basic inventory app.

- Internal Logical Files (ILF): 1 (Product Database)
- External Interface Files (EIF): 0
- External Inputs (EI): 2 (Add Product, Update Stock)
- External Outputs (EO): 1 (Low Stock Alert)
- External Inquiries (EQ): 1 (Check Price)

Cost Driver Profile: The client insists the system must be highly reusable for future branches (Factor = 5) and requires performance to be top-tier for instant scanning (Factor = 4). All other 12 factors are neutral (Factor = 3).

- Calculate the Unadjusted Function Points (UFP) using "Low" complexity weights (ILF=7, EI=3, EO=4, EQ=3).
- Calculate the Value Adjustment Factor (VAF)
- Determine the Adjusted Function Point (AFP) count.

8.

A startup is building a payment processing module.

- Data Functions: 2 ILFs (Transactions, User Profiles), 1 EIF (Bank API).
- Transactional Functions: 3 EIs (Process Payment, Refund, Change Password), 2 EOs (Monthly Statement, Fraud Report), 3 EQs (View Balance, Transaction History, Help).

Cost Driver Profile:

This system is mission critical. Use the following degrees of influence (0-5 scale):

- Data Communications: 5
- Transaction Rate: 5
- Online Data Entry: 4
- Complex Processing: 5
- Installation Ease: 2
- All others: 0
- Calculate the UFP using "Average" complexity (ILF=10, EIF=7, EI=4, EO=5, EQ=4).
- Sum the Influence Factors
- Calculate the final AFP. If 1 FP equals 8 person-hours, how many total hours are required?

9.

A multi-national shipping company needs a dashboard that pulls data from various legacy systems.

Component Type	Count	Complexity
Internal Logical Files	4	High (15 each)
External Interface Files	6	Average (10 each)
External Inputs	10	Low (3 each)

External Outputs	5	High (7 each)
External Inquiries	12	Average (4 each)

Cost Driver Profile:

The system must handle multiple sites (Factor = 5) and facilitate change for easy updates (Factor = 4). However, operational ease is a low priority (Factor = 1).

Assume the sum of the remaining 11 factors is 22.

- Calculate the total UFP based on the table provided.
- Calculate the VAF.
- Calculate the AFP.
- If the "Multiple Sites" factor dropped from 5 to 0, by what percentage would the total AFP decrease?

Numerical Exercise for PERT / CPM

1. You are estimating the time required to develop a new API. Because the legacy code is messy, your team has provided three estimates for each task.

Activity	Optimistic (to)	Most Likely (tm)	Pessimistic (tp)
A: Requirements	2 days	4 days	6 days
B: Coding	5 days	8 days	17 days
C: Testing	2 days	3 days	10 days

- Calculate the **Expected Duration** for each activity
- Calculate the **Standard Deviation** for Activity B
- What is the total expected duration for the project if these tasks are strictly sequential (A → B → C)?

2.

A marketing firm is launching a rebranding campaign. The dependencies are listed below.

Activity	Predecessor	Duration (Weeks)
A	—	3
B	A	4

C	A	2
D	B, C	5
E	C	6
F	D, E	2

- Draw the network diagram based on the predecessors.
- Identify all possible paths from Start to Finish and calculate their lengths.
- Identify the Critical Path and the total project duration.
- Calculate the Total Float (Slack) for Activity C.
- Calculate the variance for each activity